

Land Investment System - Acceptance Criteria

Land Investment, Development & Sales Management System

Acceptance Criteria

Version: 1.1 | **Companion to:** PRD v2.2 & User Flow v1.1 | **Format:** Given / When / Then, grouped by module. Each ID is stable and intended for direct use as automated test case references (e.g., in Jira/TestRail or your test suite).

AC-01 · Project (Financial Boundary)

AC-01.1 > **Given** an Admin is creating a new project > **When** they submit the form without a project name or land area > **Then** the system rejects the submission with a validation error and creates no record.

AC-01.2 > **Given** a project exists with status `CLOSED` > **When** any user attempts to add/edit an expense, payment, or investor on it > **Then** the system blocks the action and shows “Project is closed — financial edits are disabled.”

AC-01.3 > **Given** two separate projects, A and B > **When** an expense is recorded against Project A > **Then** Project B’s Actual Project Cost, cash flow, and reports are completely unaffected.

AC-02 · Landowner & Purchase Agreement

AC-02.1 > **Given** a purchase agreement with total price ₹2,00,00,000 and a 5-installment schedule > **When** the Admin records a payment of ₹50,00,000 against installment 1 > **Then** installment 1 status becomes `PAID`, and Balance Payable becomes ₹1,50,00,000.

AC-02.2 > **Given** a landowner payment installment due date has passed with no payment recorded > **When** the daily status job runs > **Then** the installment status changes to `OVERDUE` and the Landowner Dashboard reflects it.

AC-02.3 > **Given** a payment is entered with method `CASH` and amount $\geq$ ₹2,00,000 > **When** the Admin attempts to save it > **Then** the system shows a hard compliance flag referencing Section 269ST before allowing save, and logs the acknowledgement.

AC-03 · Investors & Ownership

AC-03.1 > **Given** a project with no investors yet > **When** Investor A contributes ₹50,00,000 as the only investor > **Then** Investor A’s Ownership % for this project = 100%.

AC-03.2 > **Given** Investor A holds 100% (₹50,00,000 of ₹50,00,000 total) > **When** Investor B joins with a ₹50,00,000 contribution > **Then** Total Share Capital becomes ₹1,00,00,000 and both A and B's Ownership % recalculate to 50% each, with no change to either investor's actual contributed amount.

AC-03.3 > **Given** an Admin is adding a new investor > **When** no investor agreement document is attached > **Then** the system refuses to save the `ProjectInvestor` record.

AC-03.4 > **Given** ownership method is set to `MANUAL` for a project > **When** the Admin assigns ownership percentages directly > **Then** the system warns (but does not block) if the total across all investors does not equal 100%.

AC-04 · Expenses & Actual Cost

AC-04.1 > **Given** a project with Purchase Cost ₹2,00,00,000 and no other expenses > **When** a `DEVELOPMENT` expense of ₹8,00,000 is added with the capitalization flag ON > **Then** Actual Project Cost becomes ₹2,08,00,000.

AC-04.2 > **Given** the same setup as AC-04.1 > **When** a `MARKETING` expense of ₹3,00,000 is added with the capitalization flag OFF > **Then** Actual Project Cost remains ₹2,08,00,000 (marketing tracked as a period expense, not capitalized).

AC-05 · Plots & Cost Allocation

AC-05.1 > **Given** a parent land with Actual Project Cost ₹2,27,00,000 and total allocated area 2,27,000 sq. ft, using Area-based allocation > **When** Plot #12 with area 1,000 sq. ft is created > **Then** Plot #12's allocated cost = ₹1,00,000 (i.e., ₹100/sq.ft × 1,000).

AC-05.2 > **Given** a plot with status `SOLD` > **When** any user attempts to initiate a new sale against the same plot > **Then** the system blocks it with "Plot is not available for sale."

AC-05.3 > **Given** a project's cost allocation method is changed from Area-based to Manual mid-project > **When** the change is confirmed > **Then** the system requires the Admin to review/confirm each already-created plot's new manual cost before it takes effect — no silent recalculation.

AC-06 · Sales & Buyer Installments

AC-06.1 > **Given** a sale of type `MULTIPLE_PLOTS` with 3 plots bundled for one buyer at a combined agreement amount of ₹1,00,00,000 > **When** the sale is created > **Then** exactly 3 `SaleItem` records are created, each referencing this Sale and one Plot, and each Plot's status moves to `BOOKED`.

AC-06.2 > **Given** a sale agreement of ₹30,00,000 for one installment > **When** the buyer pays ₹10,00,000 against it > **Then** installment status = `PARTIALLY_PAID`, Paid = ₹10,00,000, Remaining = ₹20,00,000 — and Project Profit calculations are **not** affected by this partial payment (profit is derived from agreement value less cost, not cash received — see AC-07.1).

AC-06.3 > **Given** a sale where sale price is below the recorded circle-rate/DLC value for that plot > **When** the Admin attempts to save the sale > **Then** the system displays a compliance flag referencing Section 43CA/50C before allowing save.

AC-06.4 > **Given** a sale at status `SALE_AGREEMENT` with no registration number or stamp-duty-paid

status recorded > **When** the Admin attempts to move it to **SOLD** > **Then** the transition is blocked until both fields are populated.

AC-06.5 > **Given** a sale is cancelled after two installments were already paid > **When** the cancellation is confirmed > **Then** the plot(s) revert to **AVAILABLE**, the sale status becomes **CANCELLED**, and the two prior payment transactions are **VOIDED** (not deleted) with a linked audit entry.

AC-07 · Profit, Loss & ROI

AC-07.1 > **Given** Net Sale Proceeds ₹2,90,00,000 and Actual Project Cost ₹2,27,00,000 > **When** Profit & Loss is calculated > **Then** Project Profit = ₹63,00,000, and this figure is independent of how much of the ₹2,90,00,000 has actually been *collected* in cash so far.

AC-07.2 > **Given** Actual Project Cost exceeds Net Sale Proceeds > **When** Profit & Loss is calculated > **Then** the system reports a Project Loss (not a negative “profit” label) and surfaces it distinctly on the dashboard.

AC-07.3 > **Given** Investor A has Ownership % 50% and Distributable Profit is ₹63,00,000 > **When** Investor Profit is calculated > **Then** Investor A Profit = ₹31,50,000, and ROI = $31,50,000 / 50,00,000 \times 100 = 63\%$.

AC-07.4 > **Given** a Profit & Loss run marked “forecast” (not Final) > **When** the Admin views Distributions > **Then** the forecast figures are visible for planning but cannot be used to create a Distribution Proposal — only a “Final” P&L run can.

AC-08 · Distribution & Settlement

AC-08.1 > **Given** Admin creates a Distribution of ₹63,00,000 across 3 investors at 50/30/20% > **When** it is saved > **Then** it is immediately recorded and available for payout — there is no approval/review status to wait on.

AC-08.2 > **Given** a distribution where ₹60,00,000 has already been paid to Investor A historically > **When** a new settlement is calculated for Investor A with total entitlement ₹81,50,000 > **Then** the remaining settlement shown = ₹21,50,000 (₹81,50,000 – ₹60,00,000).

AC-08.3 > **Given** a proposed distribution amount exceeds the project’s currently distributable profit > **When** the Admin attempts to save it > **Then** the system rejects the save with “Distribution exceeds distributable amount” — a data-integrity check, independent of any approval workflow (there is none).

AC-08.4 > **Given** a Member (not Admin) attempts to create, edit, or delete a Distribution via any API path > **Then** the request is rejected with an authorization error — Distribution management is Admin-only.

AC-09 · Security & RBAC

AC-09.1 > **Given** a user with role **MEMBER** > **When** they call any API to create, edit, or delete a Project, an Investor, a Plot, a Distribution, or Settings > **Then** the API returns an authorization error regardless of what the UI shows (server-side enforcement, not UI-only) — Member’s write access is limited to Sales and Purchases entries only.

AC-09.2 > **Given** a user with role `MEMBER` > **When** they request any Investor-related screen, list, detail, or report — by navigation or by direct API/URL access > **Then** the system returns an authorization error or empty/forbidden response; no investor name, contribution, ownership %, or distribution figure is ever returned to a Member under any request path.

AC-09.3 > **Given** a user with role `MEMBER` > **When** they view a Project's dashboard > **Then** investor-related figures (investor count, ownership %, investor capital, distributions) are omitted from the response entirely — not hidden client-side.

AC-09.4 > **Given** any financial transaction or distribution record > **When** a user without the Admin role attempts to edit or delete it > **Then** the action is blocked — Delete and Edit beyond Create/Read are Admin-only across every module (\$5/\$11 permission table).

AC-10 · Audit & Data Integrity

AC-10.1 > **Given** any financial transaction (a payment or a distribution) > **When** Admin attempts to delete it > **Then** no hard-delete option exists for that record type — only Void/Reverse, which preserves the original record and creates a linked entry. (This is a data-integrity default, not an approval gate — Admin performs it directly, with no one else's sign-off required. See PRD §9 for how to override this default if you'd rather allow hard delete here too.)

AC-10.2 > **Given** a non-financial master record with no transactions ever recorded against it (e.g., a newly created Plot with no sale) > **When** Admin deletes it > **Then** the delete succeeds outright — this is ordinary CRUD, not subject to the Void/Reverse rule in AC-10.1.

AC-10.3 > **Given** an ownership percentage is changed on an active project > **When** the change is saved > **Then** an audit log entry is created capturing old value, new value, user, timestamp, and reason — saved directly by Admin, with no separate approval step.

AC-10.4 > **Given** a value is submitted for Amount, Ownership %, Plot Area, or Sale Price > **When** the value is ≤ 0 (or Ownership % is outside 0–100) > **Then** the system rejects the submission at the API layer, independent of any client-side validation.

AC-11 · Reports & Exports

AC-11.1 > **Given** an Admin selects a Profit/Loss report for a date range with no qualifying data > **When** they export it > **Then** the system generates a valid, empty-state report file (PDF/Excel/CSV) rather than an error.

AC-11.2 > **Given** Admin exports an Investor's Statement (Investors have no login of their own — Admin generates and sends it on their behalf) > **When** the PDF is generated > **Then** Capital Contributed and Profit Earned appear as two separate, clearly labeled figures (never netted into one number).

AC-11.3 > **Given** a Member requests any investor-related report, by name or by filter > **Then** the system does not generate it — either the report type is not offered to Member at all, or the request is rejected with an authorization error.

AC-12 · Notifications (WhatsApp / SMS)

AC-12.1 > **Given** a buyer or landowner installment becomes `OVERDUE` > **When** the daily status job

runs > **Then** a WhatsApp notification is sent to the concerned party, and the item appears on the Admin/Member Overdue widget within the same run.

AC-12.2 > **Given** Admin records a Distribution (§AC-08.1 — no approval step exists) > **When** the save completes > **Then** every investor in that distribution is sent a WhatsApp notification referencing their updated statement.

AC-12.3 > **Given** the SMS channel is disabled via environment/deployment configuration > **When** any notification-triggering action occurs > **Then** no SMS is sent and no SMS-provider call is attempted, while the WhatsApp notification for the same event still sends normally — and this required no code change to configure.

AC-12.4 > **Given** the SMS channel is enabled via environment/deployment configuration > **When** a notification-triggering action occurs > **Then** an SMS is sent to the concerned party's registered number, in addition to the WhatsApp notification.

AC-12.5 > **Given** the WhatsApp or SMS provider is unreachable or returns an error > **When** a notification-triggering business action (e.g. recording a payment) is being saved > **Then** the underlying business record still saves successfully; the notification failure is logged for retry and never blocks, rolls back, or fails the transaction it was attached to.